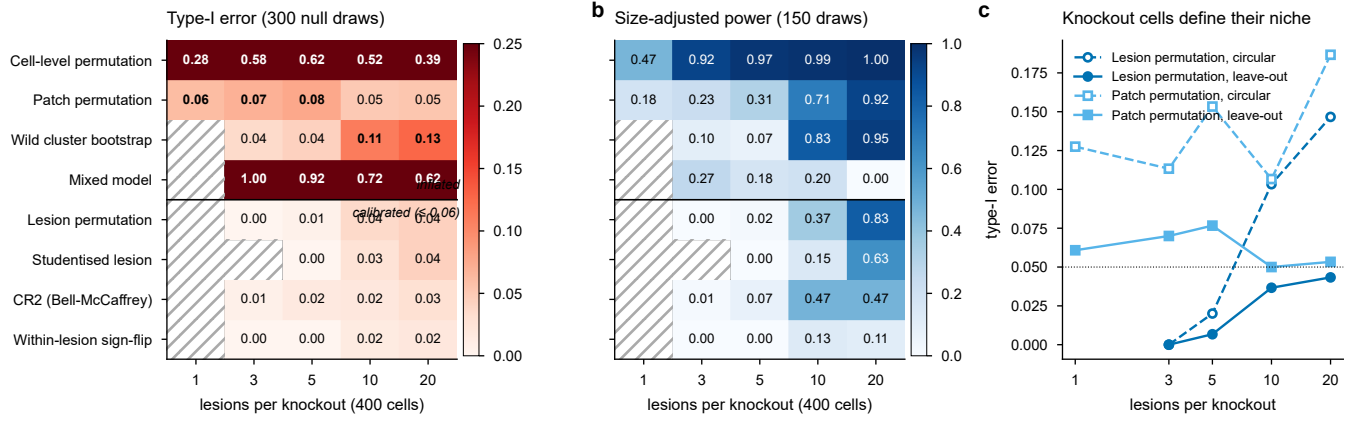

Type-I error (300 null draws)
Cell-level permutation
Patch permutation
Wild cluster bootstrap
Mixed model
Lesion permutation
Studentised lesion
CR2 (Bell-McCaffrey)
Within-lesion sign-flip
0.28 0.58 0.62 0.52 0.39
0.06 0.07 0.08 0.05 0.05
0.04 0.04 0.11 0.13
1.00 0.92 0.72 0.62
0.00 0.01 0.04 0.04
0.00 0.03 0.04
0.01 0.02 0.02 0.03
0.00 0.00 0.02 0.02
inflated
calibrated (≤ 0.06)
0.25
0.20
0.15
0.10
0.05
0.00
1 3 5 10 20
lesions per knockout (400 cells)
b
Size-adjusted power (150 draws)
0.47 0.92 0.97 0.99 1.00
0.18 0.23 0.31 0.71 0.92
0.10 0.07 0.83 0.95
0.27 0.18 0.20 0.00
0.00 0.02 0.37 0.83
0.00 0.15 0.63
0.01 0.07 0.47 0.47
0.00 0.00 0.13 0.11
1.0
0.8
0.6
0.4
0.2
0.0
1 3 5 10 20
lesions per knockout (400 cells)
c
Knockout cells define their niche
Lesion permutation, circular
Lesion permutation, leave-out
Patch permutation, circular
Patch permutation, leave-out
type-I error
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
1 3 5 10 20
lesions per knockout